On the Risks of Collecting Multidimensional Data Under Local Differential Privacy

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1. MOTIVATION & BACKGROUND

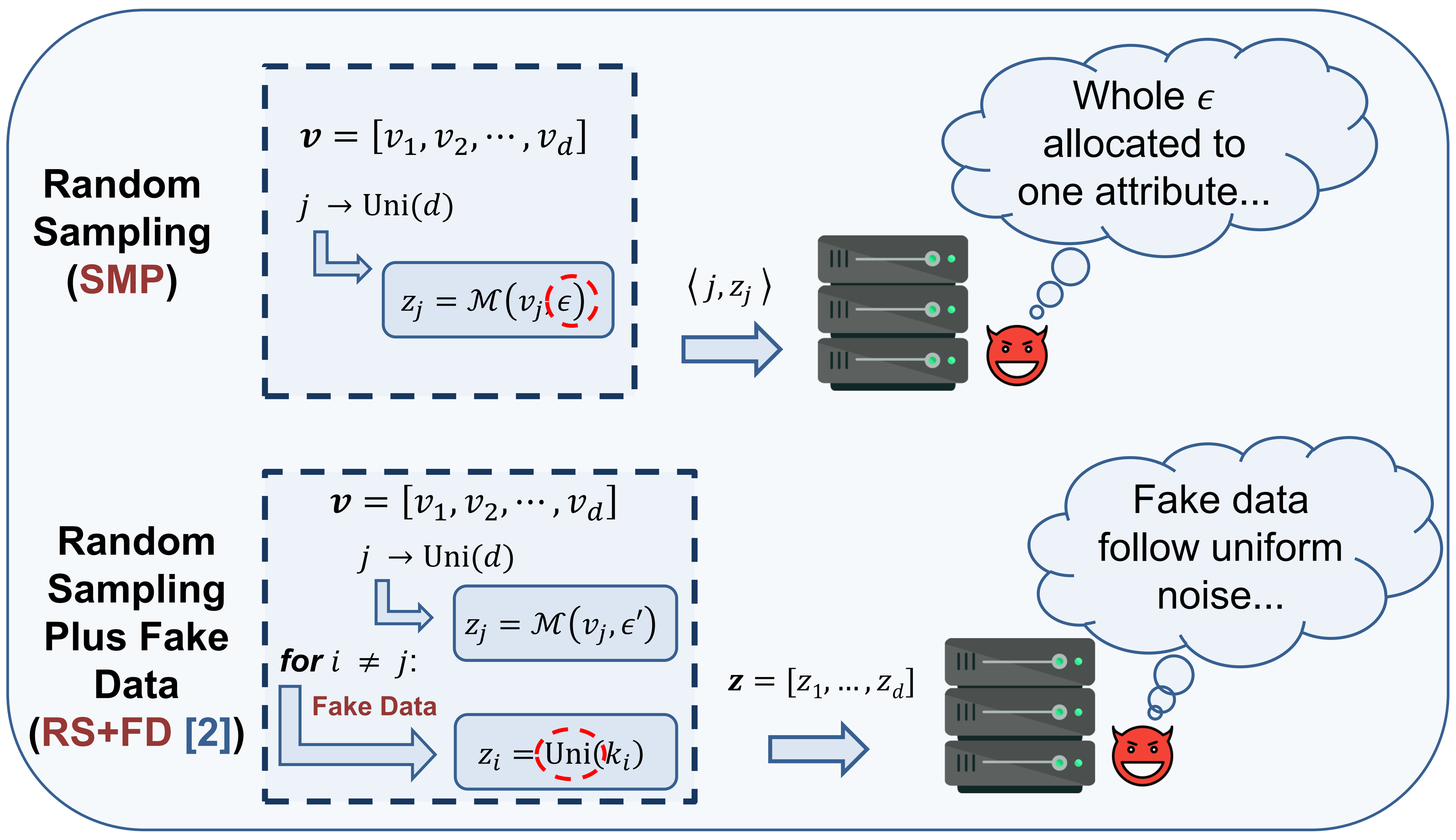
Motivations for Privacy Attacks:

- Enable vulnerability discovery.
- Allow interpreting privacy claims.
- Help practitioners to set up the privacy parameter value.

Background:

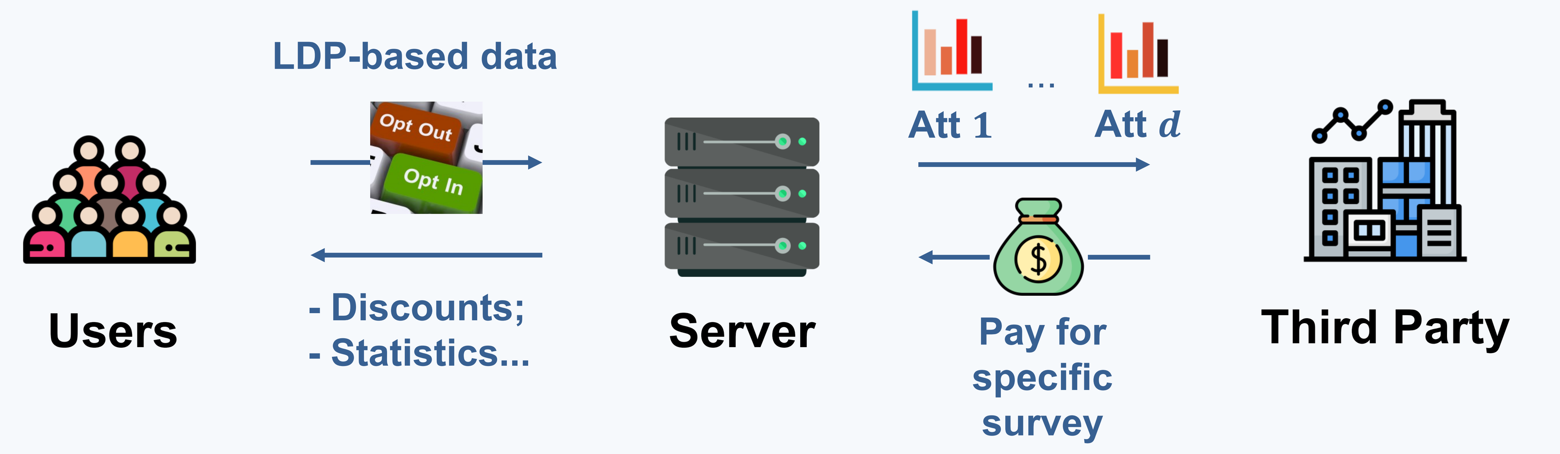
- **Privacy model:** Local Differential Privacy (LDP) protects user privacy without relying on a trusted third party [1].
- **User goal:** Report multidimensional data for $d \geq 2$ attributes $\mathbb{A} = \{A_1, \cdots, A_d\}$, for $k_i = |A_i|$, under ϵ-LDP.
- **Aggregator goal:** Estimate a k_j-bins histogram for each attribute $A_j \in \mathbb{A}$ independently.

2. LDP MECHANISMS FOR MULTIDIMENSIONAL DATA



3. TOY EXAMPLE & ASSUMPTIONS

Toy Example:

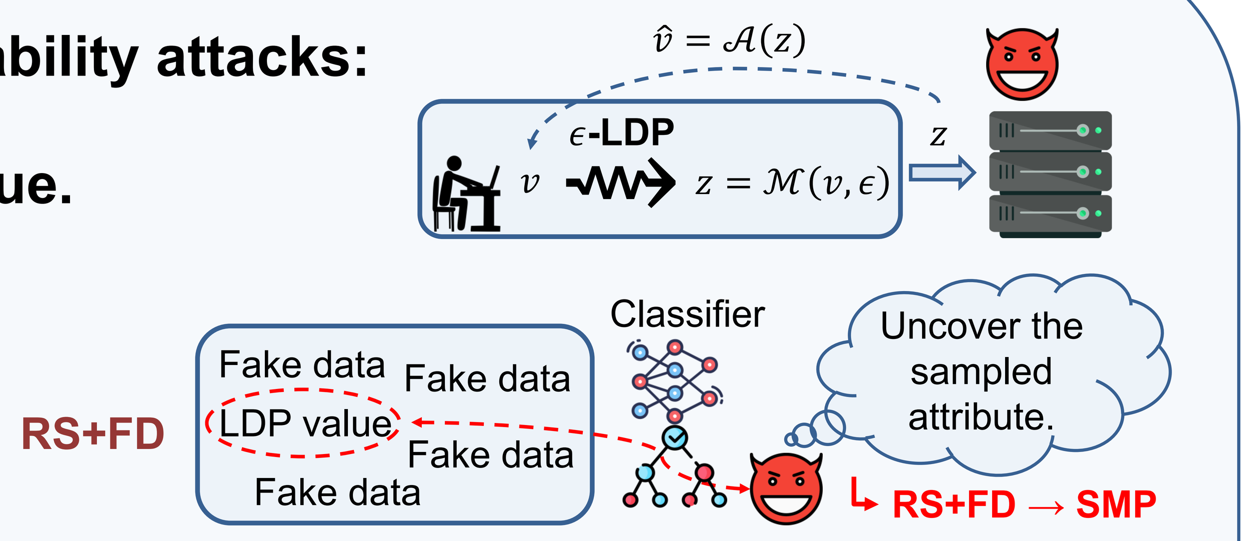


Server Assumptions:

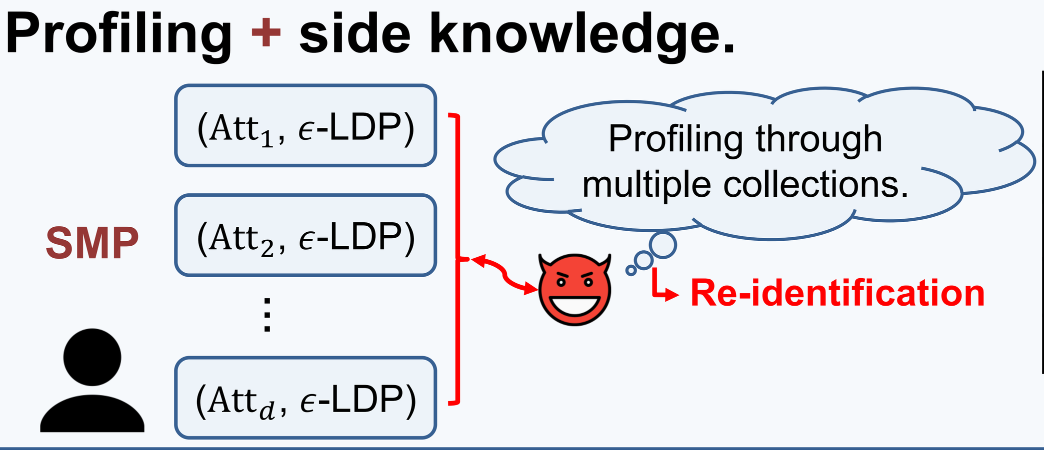
- Knows the users' pseudonymized IDs.
- Has no knowledge about the real data distributions.
- Has access to side knowledge (e.g., Census).
- Uses state-of-the-art solutions: SMP or RS+FD [2].

4. CONTRIBUTIONS (ADVERSARIAL ANALYSIS)

- **Distinguishability attacks:**
  - User's value.
  - Fake data.



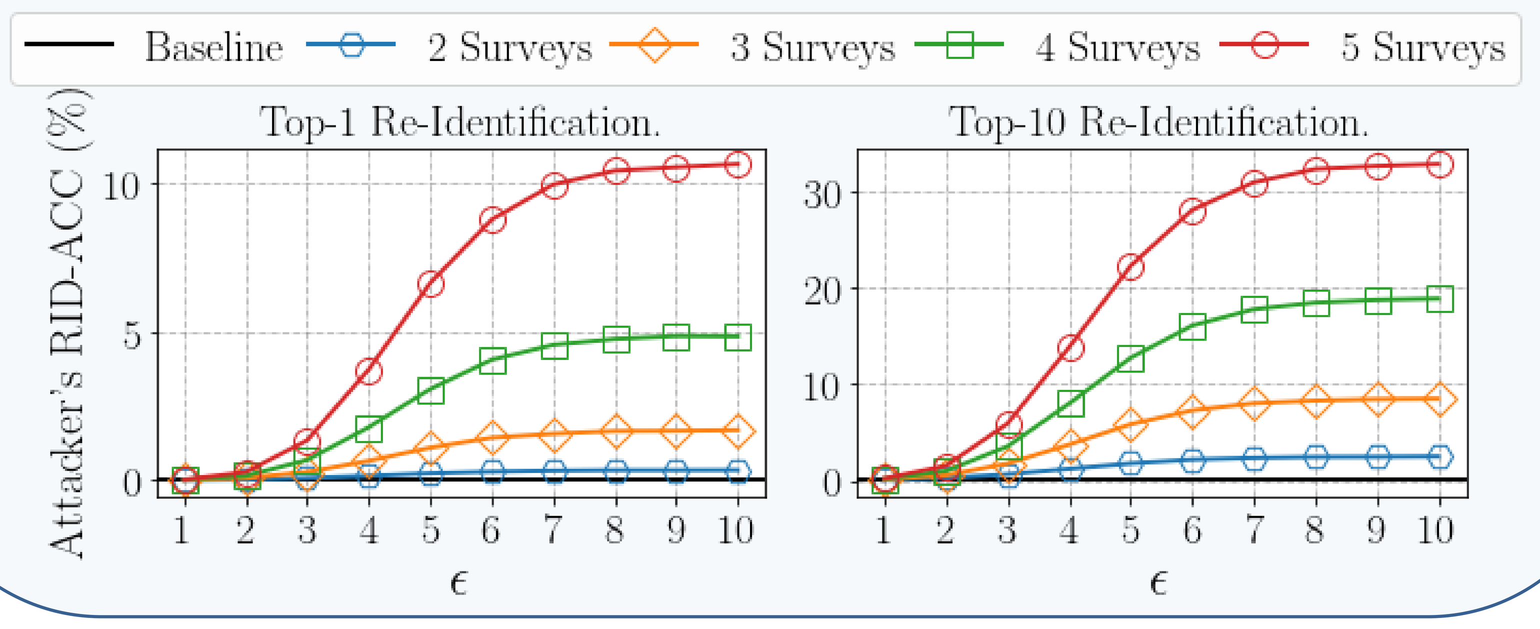
- **Re-identification attack:**
  - Profiling + side knowledge.



ID	Gender	Age	...	Att_d
1	F	25	...	...
2	F	64	...	...
⋮	⋮	⋮	⋱	⋮
n	M	30	...	...

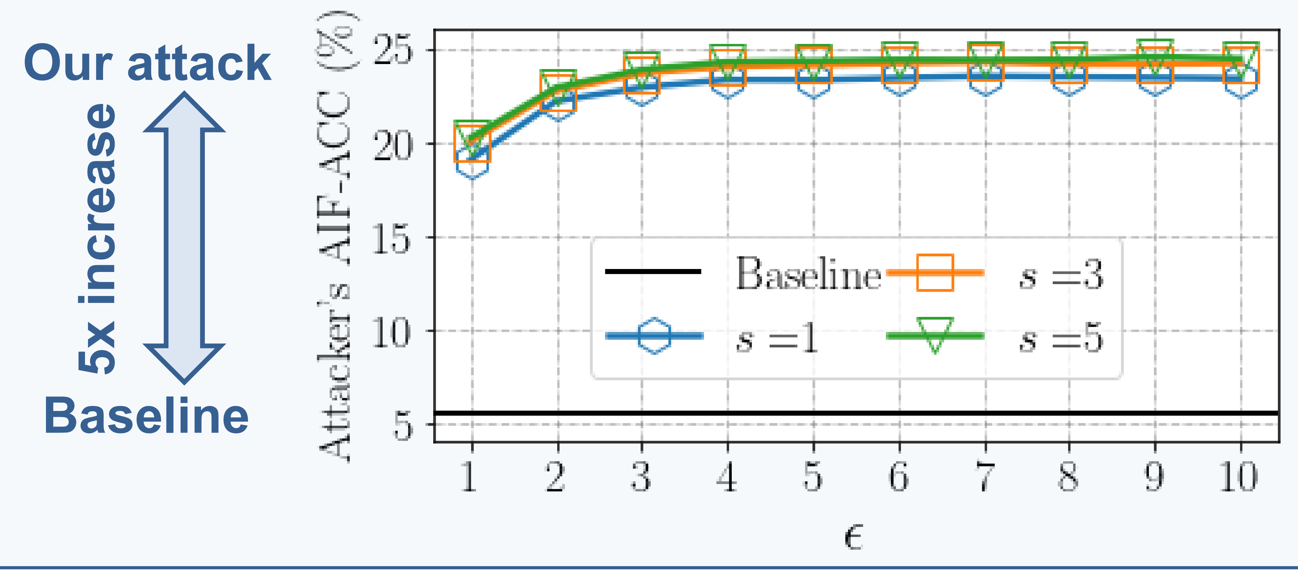
5. RESULTS ATTACKING THE SMP SOLUTION

- **Mechanism:** General Randomized Response (GRR) [3].
- **Atk goal:** Re-identify user u with top-$k \in \{1, 10\}$ guesses.
- **Metric:** Re-Identification Accuracy (RID-ACC).
- **Baseline RID-ACC:** $^{\text{Top}-k}/_{n}$, for $n = 45{,}222$.
- **Strategy:** Profiling users through $\{2, 3, 4, 5\}$ collections.



6. RESULTS ATTACKING THE RS+FD SOLUTION

- **Mechanism:** RS+FD[GRR] [2].
- **Atk goal:** Predict sampled attribute given $\boldsymbol{z} = [z_1, \dots, z_d]$.
- **Metric:** Attribute Inference Accuracy (AIF-ACC).
- **Baseline AIF-ACC:** $^{1}/_{d}$, for $d = |\mathbb{A}|$.
- **Strategy:** Adversary trains a classifier over $s \in \{1n, 3n, 5n\}$ synthetic profiles.



7. CONCLUSION

- Identified new privacy threats for LDP mechanisms.
- RS+FD → Natural countermeasure to re-identification.
- Enhanced RS+FD with realistic fake data → RS+RFD.

8. PERSPECTIVES

- Attack other more complex LDP mechanisms.
- Assess privacy risks of local d-privacy [4] mechanisms;
- Design of new countermeasure solutions.

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REFERENCES

[1] Kasiviswanathan SP, et al. "*What can we learn privately?*". In: SIAM Journal on Computing, 2011.

[2] Arcolezi HH, et al. "*Random sampling plus fake data: Multidimensional frequency estimates with local differential privacy*". In: CIKM, 2021.

[3] Kairouz P, Bonawitz K, Ramage D. "*Discrete distribution estimation under local privacy*". In: ICML, 2016.

[4] Chatzikokolakis K, et al. "*Broadening the scope of differential privacy using metrics*". In: PETS, 2013.

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